

Agenda

↳ BERT

↳ Architecture

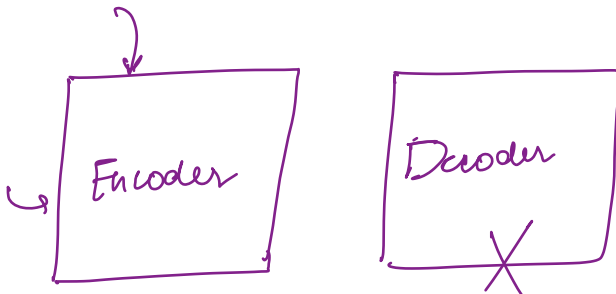
↳ Transformers

↳ Implementation ✓

↳ fine tuning

BERT : Bidirectional Encoder Representation
from Transformers (Google)

Learning representation both left → right
right → left } pretraining
Left -----> Right



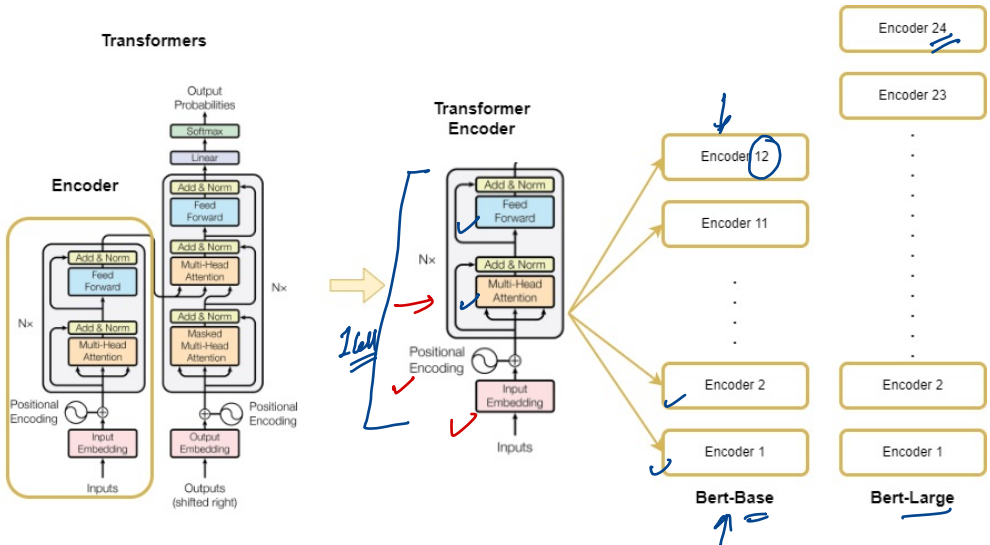
bank

rice bank

loan from bank

≈ BERT → Embeddings are better

Architecture



① Embedding model → word2vec, Glove (BERT)

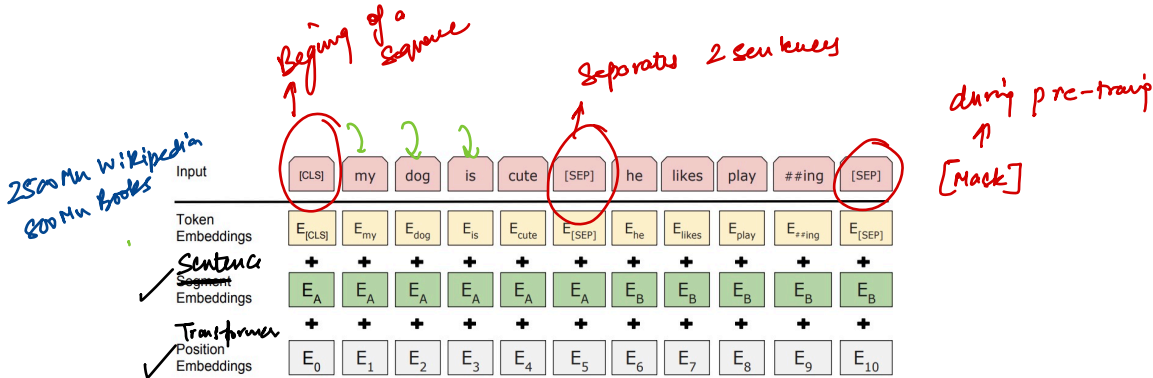
② Base Model

hidden size → 768
12 attention heads

↳ 110M Parameters

→ 1024
24 Attention heads

→ 340M Parameters



Wordpiece tokenizing

"paragliding"

para | glide | ing

paraglidicol

"para | glidi | di'col"

ing

ing

Out-of-Vocabulary (OOV) Robustness: Instead of failing on unseen words (like "unhappiness"), the model processes them as known subwords (["un", "##happi", "##ness"]). It retains context and meaning through its subword vectors.

Pre-training
+ 2 steps

Masked Language Modeling
(MLM)

★ Predicting the Masked words

[wikipedia]

[CLS] Actual sentence	Masked train sentence	Label
→ Today morning, I went for a tooth removal to my dentist	Today morning, I went for a < MASK > removal to my dentist	tooth
→ Did you go to school today	Did you < MASK > to school today	go

Semi-Supervised

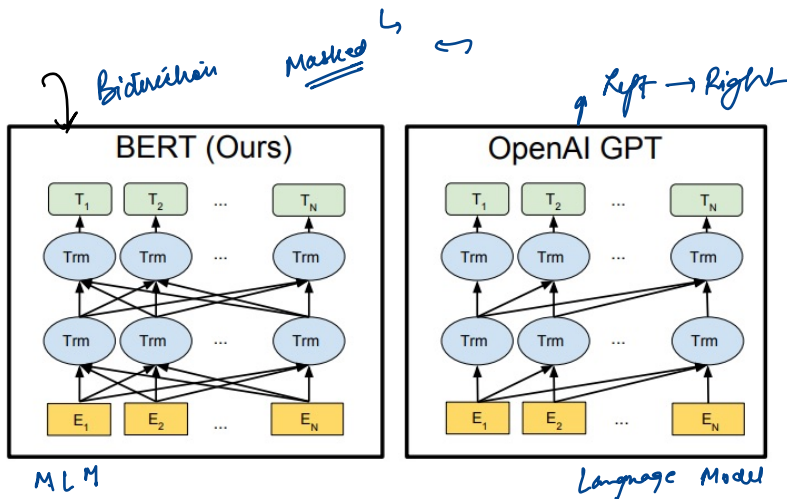
Random → 15% of the words in the corpus-

② Next Sentence Prediction (NSP)

✓ Sentence A	✓ Sentence B	IsNext?
→ The Indian cricket team won the match against West Indies.	Obama served 2 terms as the president of USA.	No
→ Nadal won the 2022 Australian open.	With this victory he now has 21 GrandSlam titles to his name.	Yes

Application

↳ question answering → NSP / text generation



Positional Encoding

Example: the word bank can represent financial bank, charging power bank or even river bank.

Let's understand it with an example

Example sentence: While travelling, I always carry a power **bank** to charge my phone and have some cash in hand as I may not find a **bank** nearby.

For simplicity, let's assume the word embedding are composed of 3 dimension, then

bank = [0.45, 0.23, 0.98]

When this embedding is used in power bank or a financial bank it may not be able to understand the contextual differences.

Contextual embedding

bank = [0.45, 0.23, 0.98] referring to power bank

bank = [0.89, 0.93, 0.62] referring to financial bank

However based on the context if the embedding is different for the same word then the model will be able to learn better.

Direct Implementation

Fine tuning or using Bert for your own use-case :

```
{
  "text": "This movie was surprisingly good and emotionally engaging.",
  "label": 1
}
```

```
{
  "text": "This movie was surprisingly good and emotionally engaging.",
  "label": 1,
  "input_ids": [101, ..., ..., ..., 102, 0, 0, 0],
  "attention_mask": [1, 1, 1, 1, 1, 1, 0, 0, 0]
}
```